

Matt Ebrahim, Ph.D.

Senior Data Scientist — Generative AI — Drug Discovery — Biotech

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Professional Summary

Ph.D.-trained AI Scientist with 5+ years designing and deploying machine learning systems across the full drug discovery pipeline — from generative molecular design through wet-lab validated candidates. Expert in GNNs, diffusion models, and multimodal transformers applied to ADMET prediction, knowledge graph-based drug repurposing, and medical imaging. Rare combination of deep ML engineering and hands-on drug discovery biology. NeurIPS-submitted researcher, U.S. patent holder, VC presenter, and graduate-level instructor at Northeastern University. Targeting Principal/Staff AI Scientist and consulting roles at AI-first biotech and pharma companies.

Professional Experience

Senior Data Scientist, Formation Bio —

Jun 2025 – Present

- Lead multiple high-impact drug repurposing initiatives, owning the full ML lifecycle from data curation and preparation through stakeholder communication and strategic decision support.
- Architected scalable graph neural network pipelines trained on large-scale biomedical knowledge graphs to estimate probability of success between drug assets and disease indications, directly informing portfolio investment decisions.
- Collaborated with data engineering and platform teams to build robust knowledge graph infrastructure and implement comprehensive data processing pipelines for GNN training at scale.
- Developed a predictive framework for M&A deal activity across 65 therapeutic indications, leveraging a dual-LLM pipeline (Opus 4.6 + Snowflake Cortex) to curate thousands of deals and engineering 69 features from DealForma data; identified statistically significant leading indicators of future deal volume, upfront value, and large pharma participation using Lasso logistic regression with temporal cross-validation (AUC-ROC up to 0.85).
- Developed MRI imaging as a surrogate endpoint for drug efficacy studies, potentially reducing clinical trial duration. Coordinated with external vendors to acquire a dataset of tens of thousands of patients with multiple MRI series and acquisitions.
- Built scalable data curation pipelines for large-scale MRI processing on GCP, including DINOv2 embedding generation, clustering-based quality assessment, and outlier removal to prepare datasets for model training.
- Designed and trained Vision Transformer (ViT) architectures for time-to-event prediction from MRI, bridging computer vision with clinical drug development applications.
- Fine-tuned domain-specific LLMs (BioMistral, BioMegatron, SapBERT) for ontology mapping framed as named-entity recognition, improving data standardization across therapeutic areas.

AI Scientist II, 1910 Genetics —

Nov 2024 – May 2025

- Led design and deployment of generative AI systems for de novo molecular generation, overseeing the full research lifecycle from literature review through production deployment.
- Developed novel multimodal AI architecture integrating molecular graph representations with molecular dynamics features for blood-brain barrier permeability prediction in CNS drug discovery (manuscript submitted to NeurIPS 2025).
- Delivered computationally designed molecules that progressed through in vitro and in vivo validation, demonstrating high target engagement with nanomolar inhibition coefficients.
- Managed cross-functional teams of data engineers; architected and deployed models on Azure ML, AWS Bedrock, and SageMaker.
- Served as regular participant in investor meetings, demonstrating AI capabilities and communicating strategic vision to stakeholders.

AI Scientist I, 1910 Genetics —

2023 – Nov 2024

- Built and deployed Graph Neural Networks, chemical language models (LSTM and Transformer architectures), and Denoising Diffusion Probabilistic Models with GNNs for de novo small molecule generation.
- Integrated multi-parameter optimization techniques (Pareto front analysis, Bayesian optimization) to identify drug candidates optimizing for novelty, potency, and metabolic stability simultaneously.
- Integrated quantum mechanics-derived features into graph models for improved ADMET prediction; benchmarked against Graphormer, AttentiveFP, and QM cross-attention embeddings.

- Developed SMILES-based RNNs with reinforcement learning fine-tuning for CNS-targeted drug discovery applications.
- Curated and standardized datasets (CrossDocked, ChEMBL, MOSES) for reproducible model development and benchmarking.

Adjunct Teaching Professor, Northeastern University –

Spring 2025 – Present

- Designed and delivered CSYE 7374: *Deep Learning and Generative AI in Healthcare*, a graduate course providing hands-on experience with state-of-the-art AI for healthcare applications.
- Developed comprehensive curriculum covering foundation models (MedGemma, BioMedCLIP), DDPMs for medical image synthesis in rare disease settings, and chemical language models for de novo drug generation.
- Created challenging programming assignments and capstone projects mirroring real-world healthcare AI problems; conducted office hours and contributed to broader university AI initiatives.
- Course site: matt-ebrahim.github.io/CSYE7374

Clinical Research Associate, Northwestern University –

2022 – 2023

- Led two major research initiatives at the intersection of medical imaging and deep learning, leveraging Northwestern's high-performance computing cluster for scalable processing.
- Developed CycleGAN architecture to synthesize MRI flow imaging from CT angiographic data, enabling hemodynamic assessment from anatomy-only scans. Implemented medical image registration algorithms for multi-modal spatial alignment.
- Developed deep learning pipelines combined with seismocardiography as a cost-effective alternative to 4D flow MRI for cardiovascular assessment, resulting in a first-author publication and U.S. patent.
- Built end-to-end CNN-MLP systems for non-invasive estimation of aortic blood flow from wearable SCG signals.

Ph.D. Research, Stony Brook University –

2017 – 2022

- Developed signal- and image-processing ML models for terahertz imaging and burn injury diagnostics.
- Published 8+ first-author papers and co-authored 30+ peer-reviewed publications.

Technical Skills

Languages: Python, C++, MATLAB, Bash

AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Optuna, PyCaret, Hugging Face

Generative Modeling: Diffusion (DDPM), Transformers (GPT, T5), VAE, GAN, RNN/LSTM

Molecular Modeling: RDKit, DeepChem, AutoDock Vina, Open Babel, ESMFold, AlphaFold

Graph ML: Graphormer, GCN, GAT, MPNN, DGL, PyG

Foundation Models: BioMistral, BioMegatron, SapBERT, MedGemma, BioMedCLIP

Cloud & MLOps: Azure ML, AWS (Bedrock, SageMaker), GCP, Snowflake, Docker, SLURM, Git, GitHub

Project Management & Collaboration: Jira, Confluence, Agile/Scrum

Domains: Drug discovery, ADMET modeling, protein-ligand design, medical imaging, knowledge graphs

Education

Ph.D., Biomedical Engineering – Stony Brook University, 2022

B.Sc., Electrical Engineering – Amirkabir University of Technology, 2016

Select Publications & Patents

- **U.S. Patent** (2025): Personalized Chest Acceleration Using Deep Learning. [Link](#)
- **Khani M.E.** et al. Multimodal Graph-Attention Networks with QM-Guided Cross-Attention for ADMET Prediction. *Submitted to NeurIPS 2025*.
- **Khani M.E.** et al. A Deep Learning Approach to Using Wearable Seismocardiography (SCG) for Diagnosing Aortic Valve Stenosis and Predicting Aortic Hemodynamics Obtained by 4D Flow MRI. *Annals of Biomedical Engineering*, 51, 2802–2811, 2023. [Link](#)
- **Khani M.E.** et al. Accurate and Early Prediction of the Wound Healing Outcome of Burn Injuries Using the Wavelet Shannon Entropy of Terahertz Time-Domain Waveforms. *Journal of Biomedical Optics*, 27(11), 116001, 2022. [Link](#)

- **Khani M.E.** et al. Supervised Machine Learning for Automatic Classification of *in vivo* Scald and Contact Burn Injuries Using the Terahertz PHASR Scanner. *Scientific Reports*, 12, 5096, 2022. [Link](#)
- **Khani M.E.** et al. Diffuse Terahertz Spectroscopy in Turbid Media Using a Wavelet-Based Bimodality Spectral Analysis. *Scientific Reports*, 11, 22804, 2021. [Link](#)
- **Khani M.E.** and Arbab M.H. Chemical Identification in the Specular and Off-Specular Rough-Surface Scattered Terahertz Spectra Using Wavelet Shrinkage. *IEEE Access*, 9, 29746–29754, 2021. [Link](#)
- **Khani M.E.**, Winebrenner D.P., and Arbab M.H. Phase Function Effects on Identification of Terahertz Spectral Signatures Using the Discrete Wavelet Transform. *IEEE Trans. Terahertz Sci. Technol.*, 10(6), 656–666, 2020. [Link](#)
- Osman O.B., Harris Z.B., Zhou J.W., **Khani M.E.** et al. In Vivo Assessment and Monitoring of Burn Wounds Using a Handheld Terahertz Hyperspectral Scanner. *Adv. Photonics Research*, 3, 2100095, 2022. [Link](#)
- Osman O.B., Harris Z.B., **Khani M.E.** et al. Deep Neural Network Classification of *in vivo* Burn Injuries with Different Etiologies Using Terahertz Time-Domain Spectral Imaging. *Biomed. Opt. Express*, 13(4), 1855–1868, 2022. [Link](#)
- 20+ additional co-authored publications and conference proceedings in terahertz spectroscopy, medical imaging, and biomedical signal processing.